

WEEKLY INTERNSHIP ACTIVITY 10

Game engines are software frameworks used to design, develop, and run video games efficiently. They provide pre-built systems such as rendering, physics, scripting, and asset management, allowing developers to focus more on game logic and creativity rather than building everything from scratch. Understanding different game engines helps in selecting the right tool based on project requirements, skill level, and long-term goals.

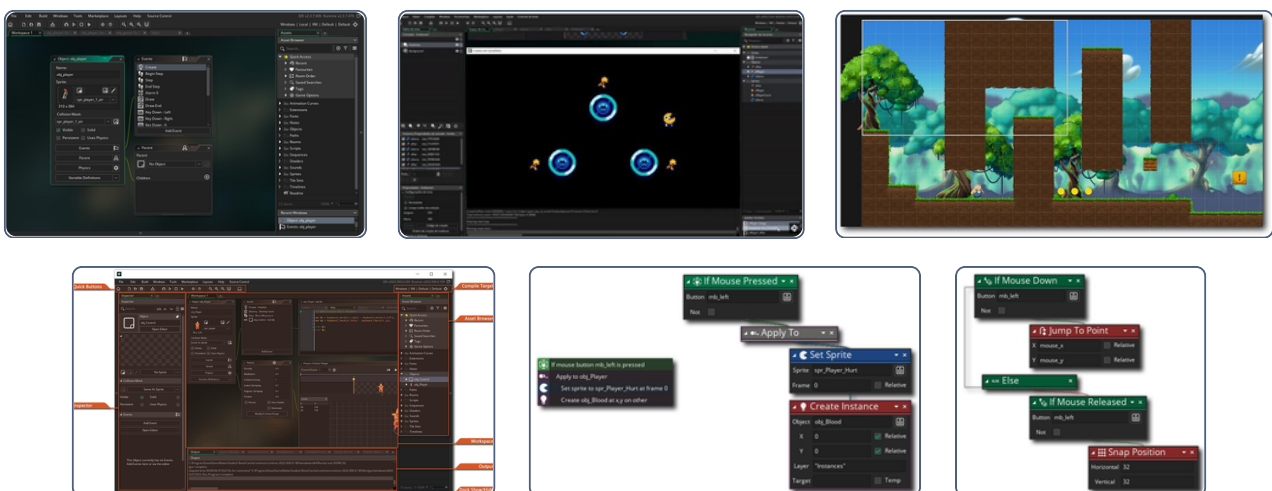
Overview of Few Selected Game Engines

UNITY



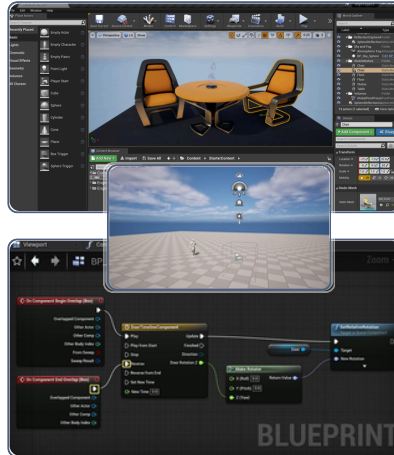
Unity is one of the most widely used game engines, known for its flexibility and strong community support. It uses C# for scripting and supports both 2D and 3D game development. It is popular among indie developers and mobile game creators due to its ease of use and cross-platform capabilities.

UNREAL ENGINE



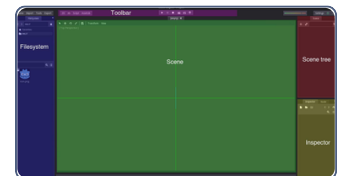
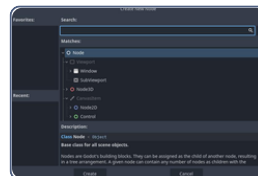
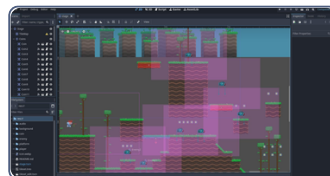
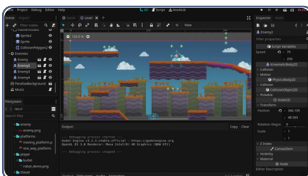
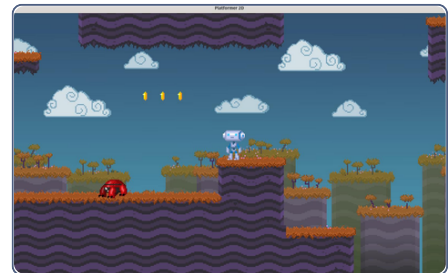
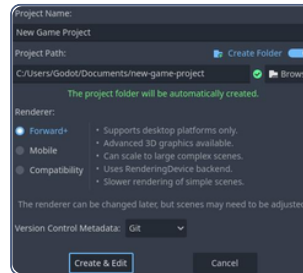
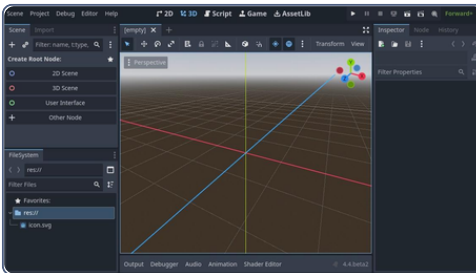
Unreal Engine is known for its high-end graphics and is commonly used in AAA game development. It uses C++ and also provides a visual scripting system called Blueprints. It is ideal for projects requiring realistic visuals and complex environments.

GODOT



Godot is an open-source game engine that is lightweight and beginner-friendly. It uses its own scripting language (GDScript) which is similar to Python. It is especially strong for 2D game development and is gaining popularity for its simplicity and flexibility.

GAMEMAKER



GameMaker is primarily focused on 2D game development and is well-known for its drag-and-drop interface. It is highly suitable for beginners with little or no coding experience and is widely used for creating simple indie games.

Comparison Table

Feature	Unity	Unreal Engine	Godot	GameMaker
Programming Language	C#	C++, Blueprints	GDScript, C#	GML, Drag-and-Drop
Best For	2D & 3D games	High-end 3D games	2D (also supports 3D)	2D games
Learning Curve	Moderate	Steep	Easy	Very Easy
Graphics Capability	Good	Excellent	Moderate	Basic
Open Source	No	No	Yes	No

Feature	Unity	Unreal Engine	Godot	GameMaker
Community Support	Very Strong	Strong	Growing	Moderate
Platform Support	Wide (PC, Mobile, Console)	Wide	Wide	Moderate

Analysis and Findings

Each engine serves a different type of developer and project scope. Unity stands out for its balance between performance, ease of use, and community support. Unreal Engine excels in delivering high-quality graphics but requires a steeper learning curve. Godot provides an open-source alternative with simplicity and flexibility, making it appealing for independent developers. GameMaker simplifies development for beginners but is limited mainly to 2D games.

Recommendation for Beginners

For beginners, Godot and GameMaker are the most suitable starting points due to their simplicity and low learning curve. Godot is ideal for those who want to learn programming concepts while still working in a manageable environment. GameMaker is best for users who prefer a visual, drag-and-drop approach with minimal coding.

However, if long-term scalability and industry relevance are priorities, starting with Unity is a strong choice. It offers a good balance between beginner accessibility and professional capabilities, making it a practical investment for future development.

Conclusion

The choice of a game engine depends on the developer's goals, experience level, and type of game being developed. While beginner-friendly engines help in understanding core concepts quickly, more advanced engines provide greater flexibility and professional opportunities. A clear understanding of these tools allows developers to make informed decisions and build efficient development workflows.